

Comparative Advantage: country can produce a good at lower cost compared to another country. not absolute advantage matters. Each country specializes in good it has relative advantage (i.e., lower OC) **MacDougall Model of Int. Capital allocation:** \$ moves from countries w. lot of \$ (& doesn't earn much) to countries with not a lot of \$ (and earns more), both countries benefit, more total world income. Hold true to MacDougall Model: Investors invest more to countries with higher rates of return, Investors invest more in countries which have less home capital. Capital flows are unidirectional. Rates of return equalize if capital flows are permitted. In reality: a) maybe holds true in reality, b,c,d DON'T **Advantages:** Intuitive and simple **Problems:** The model doesn't fully match real-world data because U.S. has most capital & major capital importer. Capital also flows in multiple directions btwn countries (cross-hauling), & the model ignores factors like risk, political stability, and infrastructure differences. **3) Currencies, Historical policies** Gold Standard = fixed \$20.67/oz gold price (1834–1914). Gold Exchange Standard failed during the Depression. Bretton Woods (1946–1971) = fixed but adjustable rates. After 1971 = mostly floating exchange rates. Euro introduced 1999–2002 with fixed rates between member countries. **b. Euro = example of OCA.** **Advantages:** more trade, lower trans. Costs, more price transparency, great economic/political integration. **Disadvantages:** Lose independent monetary policy, cannot devalue currency, vulnerable to asymmetric shocks, limited labor mobility **c. Optimal trading area:** (aka Optimal Currency Area (OCA): theory by **Robert Mundell** (won Nobel Prize), group of countries would benefit from sharing a single currency or fixed ER. **Advantages:** encourages trade, no currency fluctuations btwn member countries, lower transaction costs (no converting currencies) -> easier cross-boarder bus., greater price transparency (compare prices easier), increased economic integration, larger more efficient markets. **Disadvantages:** Loss of independent Monetary policy - countries cannot set own IR, can't adjust ER during recessions, cannot devalue currency to boost exports, Asymmetric shocks- if 1 country has recession but others don't, it can't respond independently (ex: during European crisis, countries like Greece couldn't devalue their currency b/c they used Euro). According to theory, it works best when labor is mobile btwn countries, economies are similar, are fiscal transfers (fed taxes/spending), businesses are synchronized. **4) Determinants of exchange rates:** factors that cause currency's value to rise/fall relative to another currency. 1. Inflation (PPP) higher inflation, currency tends to depreciate. Higher prices -> currency loses purchasing power -> exchange rate falls. 2. Interest Rates (IR) If country's IR rise foreign investors want higher returns, demand for that currency increase and currency appreciates. Connects to Fisher Effect and International FE. 3. Income Levels (GDP Growth) If country's economy is growing, imports may increase -> more demand for foreign currency, could cause depreciation, but strong growth can also attract investment -> appreciation. 4) Government Policies, money supply changes, fiscal policy, central bank intervention, ER regime (float vs peg). 5) Expectations/Speculation, if investors believe currency will rise they buy it now, demand increase and it appreciates immediately, ER are very forward-looking **5) Types of exchange rate regimes** Free float- gov allows exchange rates to be set by market forces. Managed float- gov may intervene through open market purchases (smoothing fluctuations, leaning against the wind). Unofficial pegging (e.g., China pegging against US\$). Target-zone agreement- countries formally agree to peg their FX rates to some band (e.g., European monetary system pre-Euro had a 2.25% leeway). Fixed exchange rates. More extreme- give up your currency and use, for instance US\$. **6) Triangular arbitrage:** you use 3 countries to take advantage of inconsistent exchange rates to make a profit. **Purchasing Power Parity (PPP):** same good should cost same amount in different places. **Absolute PPP (aka Law of One Price):** says price of goods should be exactly the same. Ab. PPP doesn't hold in reality (ex: McDonalds Big Macs). **Relative PPP:** says relative change in exchange rates should equal the difference in inflation. May hold a bit better in reality than Ab. PPP. PPP deviation decay: if exchange rate moves far from PPP value, it tends to move back toward PPP over about 4-5 yrs (mispricing shrinks slowly over time). **Problems with what PPP measures:** overall prices are usually measured by Price Indices (#s that measure how prices change over time), like CPI or Consumer Price Index (price consumer pays) and PPI (price producers receive). But, ppl in different countries don't buy same bundles of goods & people in different times don't buy the same goods (e.g., 1995- CDs, landlines 2026- streaming, smartphones) Other problems with PPP: some goods are non-traded (haircuts, real estate). Balassa-Samuelson Effect: Prices of non tradeable goods will be higher in wealthier (i.e. higher productivity) countries. Goods prices may be sticky (slow to change, ex: menu prices don't change daily, wages in fixed contracts) **b. Fisher Effect (domestic):** International Fisher Effect: real rates are assumed equal. Higher inflation country -> higher nominal IR, Lower inflation country -> lower nominal IR. Reality: doesn't hold up perfectly b/c currency risk, political risk, market frictions, different expectations ab inflation. Works better in long run for more developed countries. **c. Covered Interest Parity (CIP):** Relationship btwn Interest Rates & Exchange rates. Holds b/c of an arbitrage. **d. Unbiased forward rates (UIP):** unlike CIP, u don't lock-in all rates today. UIP doesn't hold on average. On average, countries w/ higher nominal rates: higher real rates. **Forward Contracts:** agreements to exchange currencies at pre-set time, some time in the future. Typically, with a bank. To buy a future, you need a brokerage acc. that holds initial margin requirement. Must keep the maintenance on acc. Ex: 100,000 AUS FC, initial margin= 2,090, maintenance is 1,900 (can buy/sell contract with \$2,090 in account. If value of acc. Drops below 1,900, must all money -> **Margin Call**). If you fail to maintain minimum maintenance amount, exchange would tell your broker to liquidate your position, this keeps exchange from losing money. Very large sizes are liquid, can customize size & maturity of trade. Bid-ask spread. Counterparty Risk, 90% delivered. Ex: I agree to buy 1 mil pounds (set amount) at price of 1.7200 \$/£ (set price) in 90 days (set time). Value = (St - 1.72) \$/£ x 1,000,000£ ((St - Ft) x amount) If St > 1.7200 -> you profit, If St < 1.7200 -> you lose, If St = 1.7200 -> Zero profit. **10) Options:** are an alternative to forwards & futures. Either calls or puts. Call: right to buy at preset price (strike) by a preset time (maturity date) Put: right to sell something at preset price (strike price) by a preset time (maturity date). Lingo: Maturity or EX date: last time to exercise the option. Underlying: the security which the option gives you the right to buy or sell. **European option:** Exercisable only at maturity. **American option:** exercisable at any time up to maturity. Price or premium: amount u must spend today to buy the option. Value of call @ maturity: max(S-X, 0) S= value of underlying, X= strike price) (and options on futures) **11) Money Market Hedge:** creating a artificial forward using the CIP relationship **12) Arguments for hedging:** Reduction of risk allows for better planning, reducing risk reduces probability of bankruptcy/bankruptcy costs (less risk/expected bankruptcy costs means firm can carry larger debt and therefore get more tax savings). Management can remove risk better than shareholders can b/c they know more about firm's risk than shareholders do. Easier to manage employees/plants b/c you can see their contributions better w/o noise of FX fluctuations (mute background noise of constant currency fluctuations). Smoothing earnings may reduce total tax liabilities if taxes are progressive. **against hedging:** shareholders can diversify, so firms don't have to. May be expensive (transaction costs, time, staff, etc.) Management may benefit w/o increasing shareholder wealth (agency problems). Currency markets are largely efficient. **13) Hedging transaction exposure:** protecting your firm's exchange rate changes on a known future foreign-currency flow (payables/receivables) Different Methods: Forward Contract, MMH, Futures, Options **14) Impossible Trinity:** a country can only have 2 of 3: Fixed exchange rate, free capital flows, independent monetary policy. **15) Current Account:** a country's balance of payments - trade balance (exports/imports), net income from abroad (interests, dividends), transfers (foreign aid, remittances). and the trade deficit: Imports > Exports. Current account: National Saving - Investment **17) Different types of agency problems**

Sample Question #6 fill in future values in the following table. Your company sold parts to Swiss distributor. Your firm will receive 8 million Swiss Francs (SF) in 180 days. The SF spot rate is 1.1479 \$/SF, and the 180-day forward rate is 1.1500 \$/SF. Currently you can enter into a future contract at 1.1500 \$/SF. Note, SF futures and options are for 125,000 SF per contract.

	10% depreciation in SF/\$F = 1.03311 1.1479 x (1-0.10) = 1.03311	No change in spot rates, \$/SF = 1.1479 in 180 days	10% appreciation in SF, \$/SF = 1.1479 1.1479 x (1+0.10) = 1.26269
Value of 8 million SF receivable	1.03311 x 8 mil = \$8,264,880	1.1479 x 8 mil = \$9,183,200	1.26269 x 8 mil = \$10,101,520
Value of _64_ future contracts (8 mil/125k= 64 contracts) selling = (-)	(1.1500 - 1.03311) x 8 mil = \$935,120	1.1500 - 1.1479 = 0.0021 0.0021 x 8 mil = \$16,800	1.1500 - 1.26269 = -0.11269 -0.11269 x 8 mil = -\$901,520
Sum of futures plus receivable	\$8,264,880 + \$935,120 = \$9,200,000	\$9,183,200 + \$16,800 = \$9,200,000	\$10,101,520 + (-\$901,520) = \$9,200,000
Value of _64_ (number and type) options Put (to sell)	PV(1+r)^n = 121,600 (1+0.10)^0.5 FV = \$127,535.1559 (1.1500 - 1.03311) x 8 mil - 127,535.1559 = \$807,584.8441	(1.1500 - 1.1479) x 8 mil - 127,535.1559 = -\$110,735.1559	Max(0, St - X) x amount 0(payoff)-127,500(premium cost) = -\$127,500
Sum of options receivable	807,584.8411 + 8,264,880 = \$9,072,464.841 about = 9,072,465	-110,735.1559 + 9,183,200 = 9,072,464.844 About = \$9,072,465	\$9,973,985
Money Market Hedge	\$9,335,295	\$9,335,295	\$9,335,295

Premium: 1.52 per SF -> .0152 \$/SF x 8 mil = 121,600. FV = 121,600(1.1)^0.5 = 127,500 (.5 = 180 days, 1.1 = (1 + 0.10%) 10% = rate

Part of Q6: Instead of hedging your exposures with futures, you consider hedging with options. Enter in the first column the number and type options you plan to use to fully hedge this exposure. Assume options with a 1.15 strike price maturing in 180 days currently cost 1.52 cents per Swiss franc. Further, assume that the WACC (weighted average cost of capital) of your company is 10% per year. Fill in each of the future values. What will be your total future value (including the receivable) under each scenario given this transaction? Put that in the labeled row. Does your company do better with an appreciation or depreciation if you use this hedge? Company did better with appreciation (made more money overall).

Remember: **types of options Put= Sell Call= Buy Receivable hedge: (sell) put. Payable hedge: (buy) call**

Find PV= Premium is 1.52 cents per Sf. must convert to \$. 1.52 cents/100 cents = \$0.0152 / Sf 0.0152 x 8 mil Sf = 121,600 USD (today) FV = PV x (1+r)^time in years
FV = 1261,600 (1+0.10)^0.5 (years) FV = 127,535.1559 must subtract the FV.

MMH: Forward = \$0 x (1 + rUS/1+rSF) -> 1.1500 = 1.1479 (1.05/ 1+rSF) -> 1 + rSF = 1.1479 x 1.05/ 1.1500 -> 1 + rSF = 1.04714 rSF = 4.714% (for 6 mos) PVsf= 8mil/ 1.04714 PVsf= 7,639,414 SF, 7,639,414 x 1.1479 = 8,890,757 USD. 8,890,757 x 1.05 = **\$9,335,295**

Q1: Comparative Advantage: Latvia & Estonia (each 100 units labor, both consume fish/grain) currently don't trade. Latvia: 1 fish=1 unit labor, 1 Bushel grain=2 units labor. Estonia: 1 fish=2 units labor, 1 bushel= 3 units labor. Both countries will benefit from trade, IF THERE IS TRADE: Latvia will export fish, Estonia will export grain.

Q2: Triangular Arbitrage: The Swiss Franc is trading at 1.1464 \$/Sfr, the euro is trading at 1.0828 \$/euro. If you can buy or sell Sfr/euro at 0.9451, is there an arbitrage? If so, how much can you make with one round-trip using \$1,000,000? **ANSWER:** \$1,000,000 -> 923,531.6 euro -> 872,829.7 Sfr -> 1,000,612 \$. So make \$612 per roundtrip.
Q3: Triangular Arbitrage w/ bid-ask Spreads: The South African Rand is trading 19.12 - 19.31 Rand/\$. The UK pound is trading at 1.2703-1.2712 \$/pound. What range of pound/Rand rates imply no opportunity for arbitrage? Make sure to go out to 5 decimal places in your answer. It is easier if you make all the quotes in \$/foreign currency, so 0.05178-0.05230 \$/Rand. When you do these triangular arbitrages, they always involve one bid-side and one ask-side transaction. Thus: (0.05178 \$/Rand)/(1.2712 \$/pound) = 0.04074 pound/Rand , (0.05230 \$/Rand)/(1.2703 \$/pound) = 0.04117 pound/Rand **ANSWER:** 0.04074-0.04117 pound/Rand (bid less than or equal to 0.04117 and offer more than or equal to 0.04074 pound/Rand, there is no arbitrage.

Q4: Two countries, the US and England, produce only one good, wheat. Suppose the price of wheat is \$2.50 per pound in the US, and 1.35 pounds in the UK. **Law of One (Spot Rate):** According to the law of one price, what should the \$:pound spot exchange rate be? **ANSWER:** P = 2.50/1.35 = 1.8519 \$/pound **PPP (Forward Rate/Expected Future Spot Rate):** Suppose the price of wheat over the next year is expected to rise to \$2.60 in the US and to 1.45 pounds in the UK. What should the one-year \$:pound forward rate be? **ANSWER:** P = 2.6/1.45 = 1.7931 \$/pound. **Q5:** from Shapiro] During 1995, the Mexican peso exchange rate rose from 5.33 peso/\$ to 7.64 peso/\$. At the same time, US inflation was approximately 3% in contrast to Mexican inflation of about 48.7%. **Nominal Exchange Rate Change:** a. By how much did the nominal value of the peso change during 1995? **ANSWER:** 1/5.33 = 0.1876 \$/peso, and 1/7.64 = 0.1309, or a change of (0.1309-0.1876)/0.1876 = -30.24% **Real Exchange Rate Change:** b. By how much did the real value of the peso change over this period? **ANSWER:** 1/7.64 X 1.487/1.03 = 0.189, and that is only a 0.72% change from the original rate.

Assume interest rates in the US are at 4.8% per year, and interest rates in Israel are at 3.25% per year, and that the shekel is trading at 3.76 shekels/dollar (on the spot market, i.e. for exchange today). a. If the two-year forward rate on shekels is 3.63 shekels/dollar, do you want to invest in dollars or shekels to maximize your returns for the next two years? If you invest in dollars, you get $1.048^2 - 1 = 9.8304\%$ returns. If you invest in shekels, you get $3.76 \cdot 1.0325^2 / 3.63 - 1 = 10.4235\%$ returns. So investing in shekels gives you higher returns. b. How much would you make in arbitrage profits if you can borrow \$10,000,000 or an equivalent amount of shekels over this time period? Borrow dollars, owe 10,983,040; get 11,042,346. Or a profit of \$59,305.73 c. What shekel/dollar forward rate implies no arbitrage? $F = S(1+r_{\text{home}})^2 / (1+r_{\text{foreign}})^2 = (1/3.76) \cdot 1.048^2 / 1.0325^2 = 0.274003$ \$/shekel, or 3.649601 shekel/dollar.

UIP: The current US/Mexican exchange rates are 0.05865 - 0.05872 \$/peso. Assume 2-year interest rates in the US allow you to invest at annual rates of 4.46% or borrow at 4.66%. Assume 2-year interest rates in Mexico allow you to invest at annual rates of 11.11% or borrow at 12.51%. Given uncovered interest parity, what is the range of expected exchange rates in 2 years?

UIP Equation: $\frac{E(S_t)}{S_0} = \frac{(1 + \text{interest rate})^t}{(1 + \text{interest rate}^*)^t}$ $.05865 \times (1.0446)^2 / (1.1251)^2 = .0506$ and $.05872 \times (1.0466)^2 / (1.1111)^2 = .0521$ **Answer:** .0506 - .0521 \$/peso

CIP: You are told that the current exchange rate on Bitcoin is 43,355-43,363. Assume that the 12-month investment and borrowing rates in the US are 4.8% and 5.0% (annual) respectively. If the one year forward rate on Bitcoin is 45,460 – 45,502, what range of interest rates in Bitcoin investments does that imply?

CIP Equation: Forward Rate $= S_0 \times \frac{(1 + \text{interest rate})^t}{(1 + \text{interest rate}^*)^t}$ $45460 = 43363 \times (1 + 0.05) / (1 + r^*)$ $\rightarrow 4560 \times (1 + r^*) = 43363 \times (1 + 0.05) \rightarrow r^* = 0.157\%$, $45502 = 43355 \times (1 + 0.048) / (1 + r^*)$, $r^* = -0.14496\%$ **Range= -0.14496% to 0.157%**

If 7-year rates in the US are 4.16%, and 7-year rates in Japan are 0.45%, and the current exchange rate is 148.15 Yen/\$, what exchange rate would you expect in 7-years?

Expected Future Exchange Rate (Relative PPP): $E(S_t) = S_0 \times \left(\frac{1 + \pi^*}{1 + \pi} \right)^t$ (foreign on top if looking for foreign expected, domestic on top if looking for domestic) **ANSWER:** $148.15 \text{ yen}/\$ \times (1.0045 / 1.0416)^7 = 114.932911 \text{ yen}/\$$ (about 115 yen/\$)

If you see:

If Question Gives	Use
Inflation + Prices	PPP
Interest rates + Inflation	Fisher Effect
Interest rates + Exchange rates	Interest Rate Parity

expected).

Fisher effect: $1 + r + (1 + a)(1 + \pi)$ r = nominal interest rate a = real interest rate π = inflation

$\frac{1+r}{1+r^*} = \frac{1+i}{1+i^*}$ $\rightarrow$ International Fisher Effect (IFE) Use when question gives, IR, Inflation Rates, Expected future exchange rates, used to predict how exchange rates will change based on interest differences.

Forwards: Fixed price (F), Fixed Time (T), Fixed amount. Long Forward (you agree to buy foreign currency) Value= $(S_t - F) \times \text{Amount}$. Spot rises $\rightarrow$ you gain. Spot falls $\rightarrow$ you lose.

Short Forward (you agree to sell foreign currency. Value= $(S_t - F) \times (-\text{Amount})$. if spot falls $\rightarrow$ you gain. If spot rises $\rightarrow$ you lose.

Call Option value at maturity Call= $\max(S_t - X, 0)$. Profit = $\max(S_t - X, 0) \times \text{amount} - \text{FV}(\text{premium})$. Put option value at Maturity: Put= $\max(X - S_t, 0)$. Profit= $\max(X - S_t, 0) \times \text{amount} - \text{FV}(\text{premium})$